Wedo – Ride-Hailing Platform
Implementation Plan (Phase 1)

Project Structure

```
Main
├── backend/          # Node.js + Express + TypeScript API
│   └── src/
│       ├── config/       # DB + env config
│       ├── controllers/  # auth, trip, wallet, zone, admin, whatsapp
│       ├── middlewares/  # JWT auth guard
│       ├── models/       # User, Trip, Zone, Pricing, Transaction,
Complaint, WalletTopUp
│       ├── routes/       # All REST route files
│       ├── services/     # matching.service, proxyPhone.service
│       └── sockets/      # Socket.io event handlers
├── mobile/           # React Native (Expo SDK 52) App
│   └── src/
│       ├── components/   # Reusable UI components
│       ├── config/       # API base URL, constants
│       ├── constants/    # colors, fonts, theme tokens
│       ├── i18n.ts       # English + Arabic RTL translations (full)
│       ├── navigation/   # AppNavigator (Stack)
│       ├── screens/
│       │   ├── Auth/     # Welcome, Login, SignUp, Profile, Demo,
FastIntro
│       │   ├── User/     # UserHome, ZoneSelect, Searching, TripStatus,
TripDetails, TripComplete, TripHistory
│       │   └── Driver/   # DriverHome, DriverWallet
│       ├── services/     # api.ts, socket.ts, soundManager, demoGuard
│       └── store/        # useAuthStore, useTripStore (Zustand)
├── admin/            # React.js + Vite Admin Dashboard
│   └── src/
│       ├── components/   # Sidebar, charts, shared UI
│       ├── pages/        # Overview, Drivers, Users, Trips, Zones,
Wallet, Staff, Complaints, Notifications, Transactions, DirectTopup,
DriverApproval, WalletApprovals
│       ├── services/     # API client calls
│       └── i18n.ts       # Admin localization
└── docker/           # (Phase 1D) Docker & Nginx configs
```

Tech Stack

Layer	Technology
Mobile App	React Native (Expo SDK 52)
Admin Panel	React.js + Vite + TailwindCSS + Recharts
Backend	Node.js + Express + TypeScript
Database	MongoDB
Real-time	Socket.io (WebSockets)
Caching & Queuing	Redis (matching algorithm + trip broadcasting)
State Management	Zustand (mobile), Context API (admin)
Auth	JWT (role-based: rider / driver / admin / staff)
i18n	i18next (English + Arabic RTL, both apps)
Fonts	Cairo (Arabic), Inter (English) via expo-google-fonts
Maps	react-native-maps
Deployment	VPS (Contabo) + PM2 + Nginx + SSL (Let's Encrypt)
Build	Expo EAS Build (APK for Android)

Build Phases

Phase 1A - Backend API Complete

- [x] Project scaffolding — Express + TypeScript + Mongoose
- [x] MongoDB connection (`config/db.ts`)
- [x] Socket.io server setup with CORS
- [x] Redis integration for driver matching queue
- [x] **Auth** — OTP-less JWT login/register (`auth.controller.ts`)
- [x] **User model** — Rider & Driver roles, wallet balance, vehicle info, is-online flag
- [x] **Trip model** — pickup/dropoff pins, fare, status lifecycle, rating, commission
- [x] **Zone model** — Pricing zones with coordinates for Haversine distance calc
- [x] **Pricing model** — Base fare, per-km, per-minute, min fare config
- [x] **Wallet model** — Top-up requests, transactions, commission deductions
- [x] **Trip CRUD** — create, accept, start, complete, cancel endpoints
- [x] **Wallet endpoints** — top-up request, admin approval, deduct commission
- [x] **Admin endpoints** — driver approval, user management, staff management, stats
- [x] **WhatsApp proxy** — `whatsapp.controller.ts` for support link generation
- [x] **Matching service** — Redis-backed driver geo-broadcast in radius
- [x] **Phone proxy service** — masked calling via backend relay
- [x] **JWT auth middleware** — role-based protection on all admin/driver routes

Phase 1B - Mobile App (Expo) Largely Complete

Auth Flow
- [x] `WelcomeScreen` — branded entry screen (logo + language selection)
- [x] `FastIntroScreen` — quick feature highlight slides
- [x] `DemoScreen` — guest/demo mode for stores
- [x] `LoginScreen` — email/password + Remember Me persistence
- [x] `SignUpScreen` — rider/driver registration with vehicle detail fields
- [x] `ProfileScreen` — view/edit profile, logout

User (Rider) Flow
- [x] `UserHomeScreen` — live map, pickup/dropoff pin drops, fare estimate, zone-based pricing
- [x] `ZoneSelectScreen` — manual zone override fallback
- [x] `SearchingScreen` — animated waiting state, timeout handling
- [x] `TripStatusScreen` — live driver tracking, ETA, trip timeline, SOS modal, ride share link
- [x] `TripDetailsScreen` — trip breakdown, fare, driver info, rating
- [x] `TripCompleteScreen` — summary, star rating, app store review prompt
- [x] `TripHistoryScreen` — paginated past trips

Driver (Captain) Flow
- [x] `DriverHomeScreen` — incoming trip cards (countdown timer), go-online toggle, heatmap layer, earnings summary
- [x] `DriverWalletScreen` — balance, top-up requests, transaction history, commission deduction display

Core Infrastructure
- [x] `AppNavigator.tsx` — stack navigator with role-based routing
- [x] `useAuthStore.ts` (Zustand) — auth state + AsyncStorage persistence
- [x] `useTripStore.ts` (Zustand) — active trip + driver tracking state
- [x] `socket.ts` — Socket.io client connection management
- [x] `soundManager.ts` — trip notification sound alerts
- [x] `api.ts` — Axios instance with JWT interceptor
- [x] `demoGuard.ts` — demo mode permission fence
- [x] `i18n.ts` — full EN/AR translations, RTL layout switching, Cairo/Inter font auto-select

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Phase 1C - Admin Dashboard Largely Complete

- [x] **Login** — Admin/Staff JWT login
- [x] **Overview** — KPI cards (trips, revenue, active drivers, users)
- [x] **Drivers Page** — list, search, approve/reject, view profile
- [x] **Driver Approval** — dedicated new-driver onboarding queue
- [x] **Users Page** — all riders, search, status, wallet info
- [x] **Trips Page** — live and historical trip table
- [x] **Zone Pricing** — CRUD for zones + dynamic pricing parameters
- [x] **Wallet Approvals** — top-up requests approval workflow
- [x] **Direct Top-up** — admin manually credits a driver wallet
- [x] **Transactions Page** — full financial ledger
- [x] **Staff Page** — create/manage sub-admin staff accounts
- [x] **Complaints Page** — rider/driver complaint ticket view

- [x] **Notifications Page** — broadcast push notifications (future)
- [x] **i18n (admin)** — Arabic + English

Phase 1D – Deployment ⚙️ In Progress

- [ ] Docker Compose — backend + MongoDB + Redis containers
- [ ] Nginx reverse proxy config (SSL termination, `/api` proxy)
- [ ] SSL via Certbot / Let's Encrypt
- [ ] PM2 process manager config for zero-downtime restarts
- [ ] `.env` secrets management on Contabo VPS
- [ ] Expo EAS Build — production APK signed for Google Play
- [ ] Google Play listing (store assets, screenshots, privacy policy)

> Reference docs already written: `Backend_Deployment_Guide.md`, `contabo_deployment_brief.md`, `deploy_to_server_live.md`, `store_publication_guide.md`

🔌 API Design

Base URL: `https://api.wedo.app/api`

Auth

Endpoint	Method	Description
`/auth/register`	POST	Register rider or driver
`/auth/login`	POST	Login (returns JWT)
`/auth/me`	GET	Get current user profile

Trips

Endpoint	Method	Description
`/trip/estimate`	POST	Estimate fare (Haversine + zone pricing)
`/trip/create`	POST	Create new trip request
`/trip/accept/:id`	POST	Driver accepts trip
`/trip/start/:id`	POST	Driver starts trip
`/trip/complete/:id`	POST	Driver completes trip (cash entry)
`/trip/cancel/:id`	POST	Cancel trip (rider or driver)
`/trip/history`	GET	Rider's past trips
`/trip/active`	GET	Current active trip

Zones & Pricing

Endpoint	Method	Description
`/zones`	GET	List all zones
`/zones`	POST	Create zone (admin)
`/zones/:id`	PUT	Update zone (admin)
`/zones/pricing`	GET	Current pricing config
`/zones/pricing`	PUT	Update pricing (admin)

Wallet

Endpoint	Method	Description
`/wallet/balance`	GET	Driver wallet balance
`/wallet/topup/request`	POST	Request a wallet top-up
`/wallet/topup/approve/:id`	POST	Admin approves top-up
`/wallet/transactions`	GET	Transaction history

Admin

Endpoint	Method	Description
`/admin/drivers`	GET	List all drivers
`/admin/drivers/:id/approve`	PUT	Approve/reject driver
`/admin/users`	GET	List all users
`/admin/trips`	GET	All trips (admin view)
`/admin/staff`	GET/POST	Manage staff accounts
`/admin/stats`	GET	Dashboard KPIs
`/admin/topup/direct`	POST	Direct wallet top-up
`/admin/complaints`	GET	All complaints
`/admin/notify`	POST	Send push notification

WhatsApp Support

Endpoint	Method	Description
`/whatsapp/link`	GET	Generate support deep link

Real-Time Socket Events

Event	Direction	Description
`driver:online`	Client → Server	Driver marks as available
`driver:offline`	Client → Server	Driver goes offline
`driver:location`	Client → Server	Driver GPS broadcast (every 3s)
`trip:new`	Server → Driver	New trip request card
`trip:accepted`	Server → Rider	Driver accepted notification
`trip:started`	Server → Rider	Trip in progress
`trip:completed`	Server → Rider	Trip finished
`trip:cancelled`	Server → Both	Cancellation notification
`trip:timeout`	Server → Driver	No driver found in radius

Design System

Colors

Token	Value	Usage
Primary Blue	`#1A6FFF`	Buttons, accents, active states
Dark BG	`#0A0F1C`	Screen backgrounds
Card BG	`#131929`	Cards, bottom sheets
Success	`#22C55E`	Trip complete, online badge
Warning	`#F59E0B`	Countdown timer, pending
Danger	`#EF4444`	Cancel, SOS, error
Text Primary	`#FFFFFF`	Headings, main text
Text Secondary	`#8892A4`	Subtext, labels

Typography

Usage	Font	Size	Weight
Screens / EN	Inter	14–28px	Regular / Bold
Screens / AR	Cairo	14–28px	Regular / Bold
Price display	Inter	28px	ExtraBold
Fare estimate	Inter	20px	Bold

Key Functional Notes

Dual-Role Single App
The mobile app is a **single APK** for both Riders and Drivers. On login, the backend returns a `role` field (`rider` | `driver`) which routes the user to the appropriate home screen. No separate builds needed.

Upfront Fixed Fare Model
1. Rider drops pickup + dropoff pin on the map
2. Backend calculates distance using **Haversine formula** + zone-based pricing params
3. A fixed upfront fare is displayed before confirmation
4. On trip completion, the server compares actual elapsed time vs expected — any **waiting/route overage** is added to protect driver earnings

Weak Internet Tolerance
Once the upfront price is locked and the trip starts, live connection is **not required** for the fare meter. The backend reconciles actual vs expected at trip completion.

Commission & Wallet Logic
- Wedo commission is deducted per-trip from the driver's wallet automatically
- Driver enters cash received at completion → system calculates change
- If cash > fare: excess is deposited to driver wallet ("change-to-wallet" feature)
- Drivers must maintain a positive wallet balance to stay online

Penetration Pricing Strategy

All pricing parameters are **8–12% below the Sudanese market benchmark (Tirhal, Yango):**
- `baseFare = benchmark × 0.90`
- `perKm = benchmark × 0.90`
- `perMinute = benchmark × 0.90`
- `minFare = competitive lower bound`

Wedo compensates by charging drivers a **lower commission** than competitors — net driver earnings are higher per trip despite cheaper rider fares.

Arabic RTL
Full RTL support via `i18next` + React Native RTL API. Font auto-switches between **Inter** (EN) and **Cairo** (AR). Locale defaults to device language; user can override in-app.

Safety & SOS
Every active trip screen includes:
- 1-tap **Police emergency call** button
- Discrete complaint options: "Reckless Driving", "Vehicle Mismatch", etc.
- Live trip sharing via a web-accessible tracking link (no app needed for recipient)

Phase 2 Roadmap

Feature	Description
Bankak / Fawry Integration	Auto wallet top-up via local Sudanese banking APIs — no admin approval needed
WhatsApp Bot Dispatcher	Ride requests via WhatsApp text (leverages free MTN/Zain social bundles)
PWA Lite Version	Sub-10MB progressive web app for low-storage devices
Push Notifications (FCM)	Firebase Cloud Messaging integration for alerts outside the app
Driver Ratings	Cumulative rating, auto-flag low-rated drivers for review
Promo Codes	Discount system for user acquisition campaigns
Surge Pricing	Dynamic multiplier based on real-time demand heatmap data
iOS Distribution	App Store submission after Android market validation

Reference

Owner: Yousif Hamid
Contact: +20-1157155248
E: Yousif.hamid@gmail.com